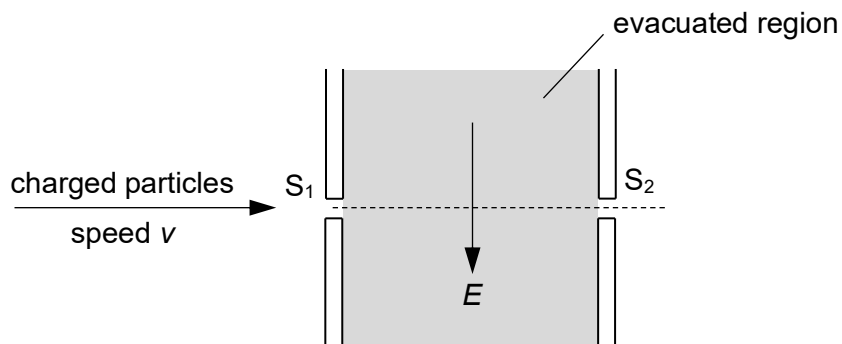


charge that negates the effect due to the two positive charges.

- 23** A narrow parallel beam of charged particles, each with speed v , passes through a slit S_1 into an evacuated region, moving in the direction towards slit S_2 .

The evacuated region is shown shaded on the diagram.



Uniform magnetic and electric fields are applied in the same evacuated region, with the electric field E in the direction shown. The particles continue to exit through slit S_2 .

What is the magnitude and direction of the magnetic field?

	magnitude	direction
A	$\frac{E}{v}$	into the plane of the paper
B	$\frac{E}{v}$	out of the plane of the paper
C	Ev	into the plane of the paper
D	Ev	out of the plane of the paper

Ans: A

$$F_B = F_E$$

$$Bqv = qE$$

$$B = \frac{E}{v}$$

24 A copper disc rotates with a constant angular velocity in a uniform magnetic field applied